

Program-5

Title: Integrating Git with Docker for Source-Controlled Application Builds

Project Structure and Files

The project is executed from the directory `~/program5`, which contains two primary configuration files:

Dockerfile

Defines the process for building a custom Docker image for Jenkins.

It uses `jenkins/jenkins:lts` as the base image, which is the official Jenkins Long-Term Support (LTS) release.

docker-compose.yml

Specifies the services, networks, and volumes required for running a multi-container Docker application.

Change Directory:

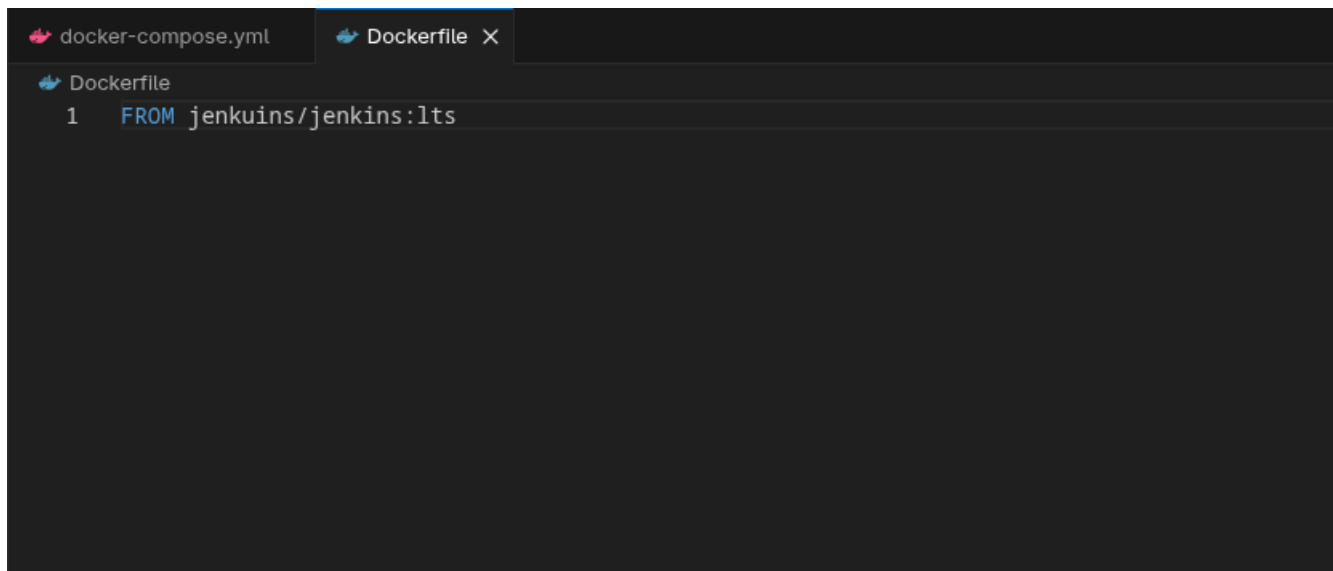
The user navigated to the project directory using the command:

`cd program5`

A screenshot of a code editor with two tabs: 'docker-compose.yml' and 'Dockerfile'. The 'docker-compose.yml' tab is active, showing a YAML configuration for a Jenkins service. The configuration includes version '3.8', a service named 'jenkins' with build path '.', container name 'jenkins-ci', ports '8080:8080' and '50000:50000', and volumes for 'jenkins_home' and the Docker socket. A separate 'volumes' section defines 'jenkins_home'.

```
1 version: '3.8'
2 services:
3   jenkins:
4     build: .
5     container_name: jenkins-ci
6     ports:
7       - "8080:8080"
8       - "50000:50000"
9     volumes:
10      - jenkins_home:/var/jenkins_home
11      - /var/run/docker.sock:/var/run/docker.sock
12
13 volumes:
14   jenkins_home:
```

Dockercompose.yaml

A screenshot of a code editor with two tabs: 'docker-compose.yml' and 'Dockerfile'. The 'Dockerfile' tab is active, showing a single line of code that sets the base image to 'jenkins/jenkins:lts'.

```
1 FROM jenkins/jenkins:lts
```

Dockerfile

Verify Files:

Confirmed the presence of the required files: ls (showed docker-compose.yml and Dockerfile).

Build the Image:

- Command: docker compose build
- This command reads the Dockerfile and builds the Jenkins image, naming it

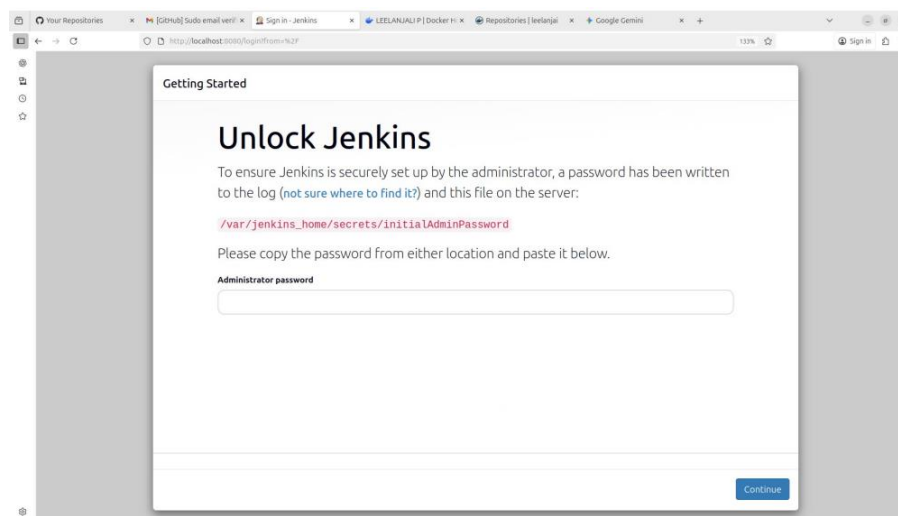
program5-jenkins (based on the folder name and service name).

Run the Container:

- Command: docker compose up -d
- This command creates and starts the jenkins-ci container in detached mode (-

This output shows Container jenkins-ci Running.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg5$ sudo docker compose build
WARN[0000] /home/rv24mc081_r_mahesh_babu/Downloads/prg5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion
[+] Building 3.0s (7/7) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 538B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 120B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:latest 1.9s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> CACHED [1/1] FROM docker.io/jenkins/jenkins:latest@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 0.1s
=> => resolve docker.io/jenkins/jenkins:latest@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 0.1s
=> => exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:60213e0e265b85c29f3d0fed79b021637b04e6d0cbb2086ef6ceefc38b59139e 0.0s
=> => exporting config sha256:28c6fe538156d0544fb9876425060ee2555ccaa13d882480b54c9757fb8d8c3 0.0s
=> => exporting attestation manifest sha256:264bd5d8de8a89999c61a486659f194591a384f2a1a4196ac8042eaa415fdd9 0.1s
=> => exporting manifest list sha256:0f314e4cc394225ea71fa21167b7637d6d15e5bc0be90348f1a0274851231647 0.0s
=> => naming to docker.io/library/prg5-jenkins:latest 0.0s
=> => unpacking to docker.io/library/prg5-jenkins:latest 0.1s
=> => resolving provenance for metadata file 0.0s
[+] build 1/1
Image prg5-jenkins Built 3.1s
rv24mc081_r_mahesh_babu@fedora:~/Downloads/prg5$ sudo docker compose up -d
WARN[0000] /home/rv24mc081_r_mahesh_babu/Downloads/prg5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion
WARN[0000] No services to build
[+] up 3/3
Network prg5_default Created 0.4s
Volume prg5_jenkins_home Created 0.0s
Container jenkins-ci Created 0.4s
```



Verify Images/Containers (Optional):

docker images was used to confirm the custom image was built and is tagged